

week-3(Pravalika-2520030402)

Question:

Write a C program to demonstrate the different states of a process (new, ready, running, waiting, and terminated) using fork(), process IDs, and process monitoring commands in Linux.

Code:

```
#include <stdio.h>

#include <unistd.h>

#include <sys/wait.h>

#include <stdlib.h>

int main()
{
    pid_t pid = fork();
    if (pid < 0)
    {
        perror("fork");
        return 1;
    }
    if (pid == 0)
    {
        printf("Child Process\n");
        printf("PID = %d\n", getpid());
        printf("PPID = %d\n", getppid());
        printf("Child is running...\n");
        sleep(10);
        printf("Child terminating...\n");
        exit(0);
    }
```

```
    else
    {
        printf("Parent Process\n");
        printf("PID = %d\n", getpid());
        printf("PPID = %d\n", getppid());
        printf("Parent waiting for child...\n");
        wait(NULL);

        printf("Child terminated.\n");
    }

    return 0;
}
```

Output:

```
rikki@BOOK-GJU4MP9458:~$ ./a.out
Child Process
Parent Process
PID = 570
PPID = 569
Child is running...
PID = 569
PPID = 326
Parent waiting for child...
Child terminating...
Child terminated.
rikki@BOOK-GJU4MP9458:~$ |
```